

THE HERITAGE SCHOOL
FIRST TERM EXAMINATION, 2026-27
PHYSICS (THEORY) · CLASS 12 · UNIT TEST 1
MARKING SCHEME

Maximum Marks: 40

Time Allotted: One and a Half Hours

Notes for the examiner

- This scheme shows **one** acceptable way to reach the answer. Award full credit for any correct alternative method, and for any equivalent wording.
- Marks against a step are for **that step**. A correct final answer reached by correct but differently ordered working earns the same total.
- Deduct $\frac{1}{2}$ mark, **once per question only**, for a numerical answer written without its unit or with a wrong unit.
- Accept answers rounded to two or three significant figures throughout; do not penalise a difference in the last figure arising from rounding partway.
- For a question answered with a diagram, the marks shown are for the **labelled** diagram. An unlabelled sketch earns no more than half the diagram marks.
- Where a question offers a choice, mark the alternative the candidate has attempted. If both are attempted and neither is struck out, mark the **first** only.
- In Section A no part marks are available: each sub-part is right or wrong.

Syllabus covered: Electrostatics; Current Electricity; Magnetic Effect of Current and Magnetism.

SECTION A - 8 MARKS

Question 1. No part marks: each sub-part scores its full mark or nothing.

Q	Expected answer	Marks
1 (i)	(b) repulsive, of magnitude kq^2/r^2 . (On contact each sphere carries $(q-3q)/2 = -q$, so both are negative.)	1
1 (ii)	(c) $R/16$	1
1 (iii)	(b) $B/2$	1
1 (iv)	(a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.	1
1 (v)	A tangential component would drive a current along the surface; in electrostatic equilibrium no current flows, so the tangential component is zero and the field is wholly normal. Accept any answer making the point that a tangential field would move the free charges.	1
1 (vi)	It decreases (to V/K). Award the mark for 'decreases' alone; the factor is not required.	1
1 (vii)	Manganin has a very low temperature coefficient of resistance, so its resistance barely changes in use. Accept also: high resistivity, so a short length suffices.	1
1 (viii)	It decreases , because raising the temperature releases many more charge carriers, so n rises much faster than τ falls. Both the direction and a reason are needed for the mark.	1

SECTION B – 10 MARKS

Question 2 [2]

$C = C_1 C_2 / (C_1 + C_2) = (4)(12)/16 = 3.0 \mu\text{F}$	$\frac{1}{2}$
$Q = CV = (3.0 \mu\text{F})(100 \text{ V}) = \mathbf{300 \mu\text{C}}$, and it is the same on each capacitor	1
$V_4 = Q/C_4 = 300/4.0 = \mathbf{75 \text{ V}}$	$\frac{1}{2}$
Total	2

Note. If a candidate finds V_4 by the ratio $V_4/V = C_2/(C_1 + C_2)$, award the same marks. Do not give credit for 300 μC stated for the series combination but then divided between the capacitors.

Question 3 [2]

$I = \mathcal{E}/r = 1.5/0.50 = \mathbf{3.0 \text{ A}}$	1
$V = \mathcal{E} - Ir = 1.5 - 1.5 = \mathbf{\text{zero}}$	1
Total	2

Note. The second mark is for **zero**, however obtained — by the equation, or by arguing that the terminals are joined by a conductor of no resistance.

Question 3 — OR [2]

Any one of: separation of the electrodes; area of the electrodes dipped in the electrolyte; concentration of the electrolyte; temperature	1
Any second factor from the same list	1
Total	2

Note. Award nothing for ‘the material of the electrodes’ alone, and nothing for factors of the *external* circuit.

Question 4 [2]

$I = V/(R_1 + R_2) = 20/10 = 2.0 \text{ A}$, the same in both	1
$V_6 = IR = (2.0)(6.0) = \mathbf{12 \text{ V}}$	1
Total	2

Note. Accept the potential-divider form $V_6 = V \times 6/(4+6) = 12 \text{ V}$ for the full 2 marks.

Question 5 [2]

Connect a high resistance in series with the galvanometer, $R = V/I_g - G$	1
$R = 10/(2.0 \times 10^{-3}) - 50 = 5000 - 50 = \mathbf{4950 \Omega}$	1
Total	2

Note. The first mark requires **in series**; a candidate who computes 4950 Ω but calls it a shunt scores 1 only. Accept $\approx 5000 \Omega$ if the working shows the 50 Ω subtracted.

Question 6 [2]

The line integral of B round any closed loop equals μ_0 times the net current threading the loop	1
$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$	1
Total	2

Note. A candidate who writes only the equation, correctly, scores 1. The words ‘closed loop’ and ‘current threading it’ are what earn the first mark.

Question 6 — OR [2]

Each coil alone gives $B_0 = \mu_0 I / 2R$ at the centre	1
The two fields are perpendicular , so they add vectorially: $B = \sqrt{2} \cdot \mu_0 I / 2R$	1
Total	2

Note. Award the second mark only if the perpendicularity is used. A candidate who adds the two fields as scalars ($2\mu_0 I / 2R$) scores 1.

SECTION C - 12 MARKS**Question 7 [3]**

Gaussian surface: a cylinder of cross-section A piercing the sheet symmetrically. By symmetry E is normal to the sheet and equal on the two flat faces; the flux through the curved surface is zero	1
$\Phi = 2EA = q/\epsilon_0 = \sigma A/\epsilon_0$	1
E = $\sigma/2\epsilon_0$, independent of distance; just outside a charged conductor E = σ/ϵ_0	1
Total	3

Note. The third mark is split: $\frac{1}{2}$ for $\sigma/2\epsilon_0$ and $\frac{1}{2}$ for the conductor result σ/ϵ_0 . A candidate who omits the factor 2 in the flux, and so obtains σ/ϵ_0 for the sheet, loses the second mark but may still earn the third if the conductor case is right.

Question 8 [3]

Junction rule: the algebraic sum of the currents at a junction is zero (conservation of charge). Loop rule: the algebraic sum of the potential differences round any closed loop is zero (conservation of energy)	1
At balance no current flows through the galvanometer , so I_1 passes through P and Q , and I_2 through R and S ; applying the loop rule to the two halves gives $I_1 P = I_2 R$ and $I_1 Q = I_2 S$	1
Dividing: P/Q = R/S	1
Total	3

Note. Both laws are needed for the first mark, $\frac{1}{2}$ each. The balance condition quoted without the no-galvanometer-current argument scores 1 of the last 2.

Question 8 — OR [3]

Parallel pair: $(6)(3)/9 = 2.0 \Omega$; total resistance $2.0 + 1.0 = 3.0 \Omega$	1
$I = \epsilon/R_{\text{total}} = 12/3.0 = 4.0 \text{ A}$; p.d. across the pair = $(4.0)(2.0) = 8.0 \text{ V}$	1
$I_6 = 8.0/6.0 = 1.3 \text{ A}$ and $I_3 = 8.0/3.0 = 2.7 \text{ A}$	1
Total	3

Note. Accept 1.33 A and 2.67 A. A candidate who forgets the internal resistance gets $I = 6.0 \text{ A}$ and loses the second and third marks; if the *method* for splitting the current is then correct, award $\frac{1}{2}$ of the third.

Question 9 [3]

Circuit diagram: driver cell with key and rheostat in series with the potentiometer wire AB; the two cells ϵ_1, ϵ_2 with their positive terminals joined to A through a two-way key; the other ends taken through a galvanometer to the jockey	2
$\epsilon_1/\epsilon_2 = l_1/l_2$, where l_1, l_2 are the balancing lengths for the two cells	1
Total	3

Note. The diagram marks are $\frac{1}{2}$ for the driver circuit, $\frac{1}{2}$ for the two-way key, $\frac{1}{2}$ for the galvanometer and jockey, and $\frac{1}{2}$ for correct polarity (like terminals joined at A). A diagram with the cells opposed cannot balance and loses the polarity $\frac{1}{2}$.

Question 10 [3]

The first wire produces at the second a field $B_1 = \mu_0 I_1 / 2\pi r$, perpendicular to it	1
A length l of the second wire feels $F = B_1 I_2 l$, so $F/l = \mu_0 I_1 I_2 / 2\pi r$ — attractive for parallel currents	1
Definition: one ampere is that steady current which, in each of two infinitely long parallel conductors of negligible cross-section 1 m apart in vacuum, produces a force of 2×10^{-7} N per metre of length	1
Total	3

Note. The direction (attractive / repulsive) is worth $\frac{1}{2}$ of the second mark. In the definition, all of '1 m apart', 'in vacuum' and the figure 2×10^{-7} N m⁻¹ are required for the full mark; omitting one costs $\frac{1}{2}$.

SECTION D - 10 MARKS

Question 11 [5]

(a) The field is $E_0 = \sigma/\epsilon_0$ across the air gap of thickness $(d-t)$ and is reduced to E_0/K inside the slab	1
$V = E_0(d-t) + (E_0/K)t = E_0[(d-t) + t/K]$	1
With $C = Q/V$ and $E_0 = Q/\epsilon_0 A$: $C = \epsilon_0 A/[d - t(1 - 1/K)]$	1
(b) Halving d doubles C , and filling with $K = 5.0$ multiplies it by five: $C' = 2KC$	1
$C' = (2)(5.0)(8.0) = \mathbf{80 \text{ pF}}$	1
Total	5

Note. In (a), a candidate who checks the result by putting $t = d$ or $t = 0$ has earned nothing extra but has done the right thing. In (b), award the fourth mark for recognising $C \propto K/d$ even if the arithmetic then slips.

Question 11 — OR [5]

(a) The torque holding the dipole at θ is $\tau = pE \sin \theta$, so $dW = pE \sin \theta d\theta$	1
Integrating from the reference position $\theta = 90^\circ$, where U is taken as zero: $U = -pE \cos \theta = -p \cdot E$	1
U is least ($-pE$) at $\theta = 0$ — stable ; greatest ($+pE$) at $\theta = 180^\circ$ — unstable	1
(b) $W = U_{180^\circ} - U_{0^\circ} = pE - (-pE) = 2pE$	1
$W = 2(2.0 \times 10^{-9})(3.0 \times 10^5) = \mathbf{1.2 \times 10^{-3} \text{ J}}$	1
Total	5

Note. The third mark is $\frac{1}{2}$ for each equilibrium position, and requires the words stable and unstable. In (b) a candidate who writes $W = pE$ (forgetting the factor 2) loses the fourth mark but keeps the fifth if the arithmetic matches their own expression.

Question 12 [5]

(a) Ampèrian loop: a rectangle with one long side of length L inside the solenoid, parallel to the axis, the opposite side well outside	1
Outside a long solenoid the field is negligible, and on the two short sides $\mathbf{B} \perp d\mathbf{l}$, so only the inner side contributes: $\oint \mathbf{B} \cdot d\mathbf{l} = BL$	1
Current threading the loop = nLI , so $BL = \mu_0 nLI$, giving $\mathbf{B} = \mu_0 n\mathbf{l}$ — independent of position, i.e. the field inside is uniform	1
(b) $n = 500/0.50 = 1000 \text{ turns m}^{-1}$	1
$B = (4\pi \times 10^{-7})(1000)(2.0) = 2.5 \times 10^{-3} \text{ T}$	1
Total	5

Note. A candidate who uses $N = 500$ in place of n and obtains $1.3 \times 10^0 \text{ T}$ loses the fourth mark; the fifth is then available only if the substitution is consistent with their own n . Accept $2.51 \times 10^{-3} \text{ T}$ or 2.5 mT .

Question 12 — OR [5]

(a) The two sides of length l perpendicular to $\mathbf{B}$ each carry a force $F = BIl$; they are equal, opposite and not collinear , so they form a couple, while the forces on the other two sides are collinear and cancel	1
Their perpendicular separation is $b \sin \theta$, where θ is the angle between the normal to the coil and $\mathbf{B}$, so $\tau = BIl \cdot b \sin \theta = BIA \sin \theta$	1
For N turns $\tau = NIAB \sin \theta$, or $\tau = \mathbf{m} \times \mathbf{B}$ with $m = NIA$	1
(b) The plane of the coil is parallel to $\mathbf{B}$, so the normal is perpendicular to it: $\theta = 90^\circ$ and $\sin \theta = 1$ — the position of maximum torque	1
$\tau = (50)(3.0)(2.0 \times 10^{-2})(0.10) = 0.30 \text{ N m}$	1
Total	5

Note. The fourth mark is for recognising $\theta = 90^\circ$ from 'plane parallel to the field'. This is the standard trap: a candidate who reads $\theta = 0$ and answers zero scores the three derivation marks and nothing more.

Mark distribution over the chapters examined

Chapter	A	B	C	D	Total
Electrostatics	3	2	3	5	13
Current Electricity	3	4	6	—	13
Magnetic Effect of Current and Magnetism	2	4	3	5	14
Total	8	10	12	10	40